

NICOLE VALENTINA SANTANA

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PROFESSIONAL PROFILE

Data Science student with a strong foundation in statistical analysis, data mining, deep learning, and electrophysiological signal processing. Practical experience in optimizing backend solutions, developing machine learning models, data visualization, and big data. Advanced skills in programming, data structures, statistical models, and GenAI.

I stand out for my analytical thinking, meticulous approach to solving complex problems, and clear technical communication. I adapt responsibly to changes and consistently contribute to collaborative work, always prioritizing quality and data-driven decision-making.

WORK EXPERIENCE

AI/ML Engineer Intern, IBM Jan 2025 – Jul 2025

- Optimized backend solutions for financial applications, focusing on contract management and APIs, improving system efficiency and scalability
- Implemented automated processes for file analysis and decryption, ensuring dependency traceability and generating structured reports
- Conducted advanced analysis of transactional databases, identifying behavioral patterns and proposing data-driven solutions
- Designed dynamic dashboards in Power BI, Streamlit, and Cognos, transforming complex data into strategic insights for technical and managerial teams
- **Achievements:** Developed an automated classification engine based on GenAI (96% accuracy), reduced critical incidents through predictive analysis, and created an AIOps chatbot for automated incident resolution

ACADEMIC EXPERIENCE

Researcher, Universidad Externado de Colombia Aug 2024 – Present

Influence of Brain Waves on Artificial Neural Network Learning

- Designed and compared bio-inspired architectures with standard deep learning models for image classification tasks
- Implemented experiments in Python using TensorFlow, PyTorch, NumPy, and Matplotlib
- Developed visualizations to interpret model behavior during training
- Applied statistical techniques to validate results and assess resilience to noise

Researcher, Universidad Externado de Colombia Nov 2024 – Dec 2024

Impact of Temporal Pressure on Neurophysiological Responses

- Analyzed neurophysiological and behavioral data from 40 officers using EEG and GSR signals
- Processed and analyzed data to identify correlations between brain activity and moral decision-making
- Implemented quantitative statistical analysis using sociodemographic surveys
- Developed insights on neuroethics and decision-making in high-pressure environments

Researcher, Universidad Externado de Colombia Aug 2023 – Nov 2023

Renewable Energy Prediction

- Implemented and compared multiple linear regression models using Bayesian and frequentist approaches
- Analyzed per capita renewable energy generation capacity using a global database
- Developed predictive models for evaluating sustainable energy policies

Interactive Dashboard Developer, Catch Consulting Feb 2023 – Jun 2023

- Developed an interactive dashboard to transform HR executive reports into decision-making tools

- Applied ethical visualization principles, data storytelling, and user-centered design
- Analyzed confidential information while preserving organizational privacy
- **Achievement:** Best Presentation Award

Researcher, Universidad Externado de Colombia

Feb 2023 – Apr 2023

Digital Image Manipulation Detection

- Developed a system for detecting digital image alterations using frequency domain analysis
- Implemented Fourier Transform to detect anomalous patterns and discontinuities
- Used Python, OpenCV, NumPy, and Matplotlib for spectral footprint extraction
- Achieved high accuracy in detecting subtle alterations through segmentation

EDUCATION

Bachelor's Degree in Data Science with Microcertification in Neuroscience,
Universidad Externado de Colombia

2022 – 2025

Open Year, Universidad de los Andes, Colombia

Jan 2021 – Jun 2021

Academic High School Diploma, Colegio La Merced

2020

Technical High School Program: Basic Sciences Applied to Engineering,
Universidad Sergio Arboleda

2018 – 2020

ADDITIONAL TRAINING

- **Course: Software Development From A to Z** – Udemy (2025)
- **Course: Artificial Intelligence & Machine Learning for Business** – Udemy (2025)
- **Course: Azure DevOps: Build and Deploy App using CI and CD Pipeline** – Udemy (2025)
- **Course: Monitoring with Prometheus, Grafana, Spring Boot, and Docker** – Udemy (2025)
- **DevOps Bootcamp** – IBM (2025)
- **Data Analysis and Visualization Bootcamp** – CYMETRIA (2024)
- **Seminar: Image Processing - Seeing the Invisible** – Universidad Externado de Colombia (2023)
- **Code in Place Bootcamp** – Stanford University (2023)
- **Course: Developing Skills for Innovation in the Fourth Industrial Revolution** – Universidad Nacional de Colombia (2020)

LANGUAGES

- **English:** C2
- **French:** B1

SOFTWARE SKILLS

- **Programming Languages:** Python, R, SQL
- **Machine Learning and Deep Learning:** TensorFlow, PyTorch, scikit-learn
- **Data Visualization:** Power BI, Streamlit, Cognos, Matplotlib, Seaborn
- **Statistical Analysis:** NumPy, Pandas, SciPy
- **Big Data:** Apache Spark, Hadoop
- **Signal Processing:** OpenCV, signal processing libraries
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Version Control:** Git, GitHub
- **Development:** APIs, Backend optimization

AWARDS

- **Best Presentation** – First Data Visualization Contest (Catch Consulting – Universidad Externado, Jun 2023)
- **Excellence Medal** – Colegio La Merced (Nov 2020) – Recognition for the best academic performance in the graduating class

NOTES

- For more information, visit my LinkedIn profile: [linkedin.com/in/nicolevalentina-santana](https://www.linkedin.com/in/nicolevalentina-santana)
- To view the Spanish version of the CV, click the following link: [Spanish CV](#)